

ROUND 5 – ADVANCED (C++)

1. Binary Search Implementation

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,3,5,7,9};
    int key=7;
    int low=0,high=4;
    while(low<high){
        int mid=(low+high)/2;
        if(arr[mid]==key)
            cout<<"Found";
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,3,5,7,9};
    int key=7;
    int low=0,high=4;
    bool found=false;

    while(low<=high){
        int mid=(low+high)/2;
        if(arr[mid]==key){
            found=true;
            break;
        }
        else if(arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }

    cout<<(found?"Found":"Not Found");
    return 0;
}
```

2. Linear Search Implementation

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={4,6,8,10};
    int key=8;
    for(int i=0;i<4;i++){
        if(arr[i]!=key)
            cout<<"Not Found";
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={4,6,8,10};
    int key=8;
    bool found=false;

    for(int i=0;i<4;i++){
        if(arr[i]==key){
            found=true;
            break;
        }
    }

    cout<<(found?"Found":"Not Found");
    return 0;
}
```

3. Check if Two Arrays are Equal

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int a[]={1,2,3};
    int b[]={1,2,3};
    if(a==b)
        cout<<"Equal";
    else
        cout<<"Not Equal";
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int a[]={1,2,3};
    int b[]={1,2,3};
    bool equal=true;

    for(int i=0;i<3;i++){
        if(a[i]!=b[i]){
            equal=false;
            break;
        }
    }

    cout<<(equal?"Equal":"Not Equal");
    return 0;
}
```

4. Find Longest Word in a Sentence

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
```

```

    string s="I love programming";
    cout<<s;
    return 0;
}

```

Correct Code:

```

#include <iostream>
using namespace std;

int main() {
    string s="I love programming";
    string current="", longest="";

    for(int i=0;i<=s.length();i++){
        if(s[i]!=' ' && s[i]!='\0'){
            current+=s[i];
        }
        else{
            if(current.length()>longest.length())
                longest=current;
            current="";
        }
    }

    cout<<"Longest word: "<<longest;
    return 0;
}

```

5. Longest Common Prefix

Buggy Code:

```

#include <iostream>
using namespace std;

int main() {
    string a="flower",b="flow";
    cout<<a;
    return 0;
}

```

Correct Code:

```

#include <iostream>
using namespace std;

int main() {
    string a="flower",b="flow";
    string prefix="";
    int len=min(a.length(),b.length());

    for(int i=0;i<len;i++){
        if(a[i]==b[i])
            prefix+=a[i];
        else
            break;
    }

    cout<<"Prefix: "<<prefix;
    return 0;
}

```

6. Maximum Occurring Character

Buggy Code:

```

#include <iostream>

```

```
using namespace std;

int main() {
    string s="sample text";
    cout<<s[0];
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    string s="sample text";
    int freq[256]={0};
    char res;
    int maxCount=0;

    for(int i=0;i<s.length();i++)
        freq[s[i]]++;

    for(int i=0;i<s.length();i++){
        if(freq[s[i]]>maxCount){
            maxCount=freq[s[i]];
            res=s[i];
        }
    }

    cout<<"Max occurring char: "<<res;
    return 0;
}
```

7. Bubble Sort Implementation

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={5,1,4,2};
    for(int i=0;i<4;i++)
        for(int j=0;j<4;j++)
            if(arr[i]<arr[j])
                swap(arr[i],arr[j]);
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={5,1,4,2};
    int n=4;

    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            if(arr[j]>arr[j+1]){
                int t=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=t;
            }
        }
    }

    for(int i=0;i<n;i++)
```

```

        cout<<arr[i]<<" ";
    return 0;
}

```

8. Check String Contains Only Digits

Buggy Code:

```

#include <iostream>
using namespace std;

int main() {
    string s="123a";
    if(s.length()>0)
        cout<<"Digits";
    return 0;
}

```

Correct Code:

```

#include <iostream>
using namespace std;

int main() {
    string s="12345";
    bool ok=true;

    for(int i=0;i<s.length();i++){
        if(s[i]<'0' || s[i]>'9'){
            ok=false;
            break;
        }
    }

    cout<<(ok?"Only Digits":"Contains Characters");
    return 0;
}

```

9. Decimal to Binary Conversion

Buggy Code:

```

#include <iostream>
using namespace std;

int main() {
    int n=10;
    cout<<n;
    return 0;
}

```

Correct Code:

```

#include <iostream>
using namespace std;

int main() {
    int n=10;
    int bin[32],i=0;

    while(n>0){
        bin[i++]=n%2;
        n/=2;
    }

    for(int j=i-1;j>=0;j--)
        cout<<bin[j];
}

```

```
    return 0;
}
```

10. First Repeated Element in Array

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,3,2,4};
    cout<<arr[0];
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,3,2,4};
    int n=5;
    bool found=false;

    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                cout<<"First Repeated: "<<arr[i];
                found=true;
                break;
            }
        }
        if(found) break;
    }

    if(!found)
        cout<<"No Repetition";
    return 0;
}
```